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EAT No. _____

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SARDAR PATEL UNIVERSITY
B.Sc. (CA&IT) EXAMINATION, 1st SEMESTER

26th October Friday, 2018

PS01CHT23 [Computer Organization & Digital Electronics]

Time::10:00 a.m. to 1:00 p.m.

Maximum Marks: 70

Q-1 Multiple Choice Question.[Each Question carries one Mark] [10]

- 1) _____ performs operations such as addition and Boolean and needed to carry out the instructions.
A. Arithmetic Logic Unit B. Registers
C. Control Unit D. None of above
- 2) The full form of EPROM is _____.
A. Erasable PROM B. Electronics PROM
C. Electric PROM D. Effective PROM
- 3) When there is no mechanical contact between the print head and paper then this printer is known as _____.
A. Impact printer B. Non-impact printer
C. both 1 and 2 D. None of them
- 4) The number $(100101)_2$ is equivalent to octal _____.
A. 54 B. 45
C. 37 D. 25.
- 5) Octal coding involves grouping the bits in _____.
A. 5's B. 7's
C. 4's D. 3's.
- 6) The digital systems usually operate onsystem.
A. binary B. decimal
C. octal D. hexadecimal.
- 7) An invert gate is also called a _____ gate.
A. NOR B. NOT
C. XNOR D. NAND
- 8) A combinational circuit that performs the arithmetic addition of three bits is called _____.
A. Full Adder B. Half Adder
C. Binary Adder D. Decoder
- 9) In D flip-flop, when CLK is low then input is _____.
A. High B. Low
C. Don't care D. Not change
- 10) A _____ is a combinational circuit that converts binary information from the 2^n coded inputs to a maximum of n unique outputs.
A. Half Adder B. Decoder
C. Encoder D. Comparator

(1)

(PTO)

- Q-2 Give Answers for the following:(Any ten) [20]
1. List out the limitations of Primary Storage.
 2. What are the five basic operations performed by any Computer System?
 3. Draw the classification of Secondary Storage Devices.
 4. Explain binary number system
 5. What is 3-bits Octal to Binary method? Explain.
 6. What is 4-bits Hexadecimal to Binary method? Explain.
 7. Explain NAND, OR gate.
 8. Explain XNOR, AND gate.
 9. Draw the circuit for $A'B+B'C+AC'$.
 10. Define encoder in short.
 11. Define decoder in short.
 12. Describe multiplexer in short.
- Q -3 A) Draw a block diagram of Basic Organization of a Computer System and explain the functions of the various units. [5]
- B) Explain Laser printer in detail. [5]
- OR**
- Q -3 A) Explain any one Evolution/generation of computers. [5]
- B) Explain keyboard in detail. [5]
- Q -4 Explain octal number system and convert $(10101011110)_2$ (binary number) into octal , hexadecimal and decimal number system. [10]
- OR**
- Q -4 Explain Hexadecimal number system and convert $(299)_{10}$ (decimal number) into binary, octal and hexadecimal number system. [10]
- Q -5 A) Explain de Morgan's first and second theorem. [5]
- B) Explain half adder in detail. [5]
- OR**
- Q -5 A) Explain associative and commutative law. [5]
- B) Simplify this Boolean expression and draw circuit: $ABC'+ABC+A'BC$ [5]
- Q -6 A) Explain 3x8 line decoder in detail. [5]
- B) Explain Ring counters. [5]
- OR**
- Q -6 A) Explain 4x1 multiplexer in detail. [5]
- B) Explain D flip-flop in detail. [5]

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 (2)